JAVA PRACTICAL PROGRAM REPORT

Simple Interest and Total Amount Calculator

Program Name: BankInterest.java

Language: Java

Topic: Scanner input and arithmetic operations

This report explains the supplied Java program, its formula, algorithm, class structure, source code, execution commands, and sample console output.

1. Aim

To accept the principal amount, annual rate of interest, and time in years, then calculate simple interest and the total amount.

2. Requirements

• Java Development Kit (JDK)
• Text editor or Java IDE
• Save the file as **BankInterest.java**

3. Formula Used

Simple Interest = (Principal × Rate × Time) / 100

Total Amount = Principal + Simple Interest

4. Algorithm

1. Start the program.
2. Create a Scanner object to read keyboard input.
3. Read principal, rate of interest, and time in years.
4. Calculate interest using (principal × rate × time) / 100.
5. Add interest to principal to calculate the total amount.
6. Display the interest and total amount.
7. Close the Scanner and end the program.

5. UML / Class Diagram

BankInterest + main(args: String[]): void
Read principal, rate, and time using Scanner
Calculate simple interest and total amount
Display results and close Scanner

The program has one class, BankInterest, and one main method. The method accepts three numeric inputs, performs the calculations, displays the results, and closes the Scanner.

6. Source Code

```
import java.util.Scanner;

public class BankInterest {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Principal Amount: ");
        double principal = sc.nextDouble();

        System.out.print("Enter Rate of Interest: ");
        double rate = sc.nextDouble();

        System.out.print("Enter Time in Years: ");
        double time = sc.nextDouble();

        double interest = (principal * rate * time) / 100;

        double totalAmount = principal + interest;

        System.out.println("Interest: " + interest);
        System.out.println("Total Amount: " + totalAmount);

        sc.close();
    }
}
```

7. Compilation and Execution

```
javac BankInterest.java
java BankInterest
```

Sample input values: Principal = 10000, Rate = 5, Time = 2 years.

8. Console Output

The code was compiled and run in the available Java environment. The following is the captured sample interaction; typed input is displayed beside each prompt for readability.

```
Enter Principal Amount: 10000
Enter Rate of Interest: 5
Enter Time in Years: 2
Enter Principal Amount: Enter Rate of Interest: Enter Time in Years: Interest: 1000.0
Total Amount: 11000.0
```

Calculation: Interest = (10000 × 5 × 2) / 100 = 1000.0. Total Amount = 10000 + 1000.0 = 11000.0.

9. Result / Conclusion

The program calculates simple interest and adds it to the principal to obtain the total amount. For the sample inputs, interest is 1000.0 and the total amount is 11000.0. Since the program uses double variables, decimal values may appear in the output.